
Residual-Budgeted Ansatz Growth for Quantum Dynamics on Sparse Qubit Graphs

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Abstract

Adaptive variational dynamics can produce compact logical circuits whose advantage is reduced by limited qubit connectivity. We introduce residual-budgeted frontier growth, which changes AVQDS generator selection only when a cheaper physical extension retains a prescribed fraction of the original McLachlan-residual improvement. A bounded set of persistent layouts supplies the incremental routing costs; the variational equations, canonical Hamiltonian pool, and regularized solver remain unchanged. Noiseless eight-spin Ising quenches and anisotropic XY ramps are evaluated on a square ladder and a branching heavy-hex graph, with paired initial states and adaptive time meshes. The square embedding gives unit CNOT and depth overhead. On heavy-hex, mean overhead falls from $2.435\times$ to $1.516\times$ in CNOT count and from $2.369\times$ to $1.467\times$ in depth relative to unconstrained AVQDS. Final fidelities are at least 99.96035%. Routing-only ablations isolate a further benefit from changed ansatz growth, while a depth regression and residual-budget sensitivity identify its limits. Per-step gate audits, exhaustive tied-embedding controls, and a linear-chain extension support a reproducible account of topology-induced cost.

1 Introduction

Quantum dynamics is a natural application of quantum computers: local interactions generate states that are difficult to describe classically, while the same interactions admit compact quantum circuits. This motivation connects the original proposals for quantum simulation [1, 2] with algorithms for fermions, spectral estimation, and chemistry [3, 4, 5, 6, 7]. Reviews and demonstrations have clarified both the scope of the opportunity and the substantial resource requirements [8, 9, 10, 11, 12, 13]. On devices with restricted connectivity, however, a short *logical* circuit need not be a short executable circuit. The distinction is especially consequential for adaptive algorithms, because the circuit is still being chosen when its hardware cost becomes relevant.

Adaptive variational quantum dynamics simulation (AVQDS) [65] chooses Pauli rotations along a trajectory using McLachlan’s principle. Its strength is to grow the variational manifold only when the current tangent space cannot describe the desired evolution. Yet a residual-only choice assigns the same status to an operator that is native to the device and an almost equally useful operator requiring several routing operations. Conversely, excluding all nonlocal Hamiltonian terms can remove directions needed to follow the original physical model. A connectivity constraint should therefore enter the *choice between useful tangent directions*, without changing the Hamiltonian or the equations of motion.

We investigate residual-budgeted frontier growth (RFG). At every expansion, RFG preserves a specified fraction of the residual improvement offered by the original AVQDS choice, and uses that allowance to reduce the additional SWAP cost of the growing ansatz. A small frontier of alternative physical layouts carries routing information between expansions. This is a concrete form of joint ansatz construction and routing: changing a selected generator changes later tangent spaces, while

changing a layout changes the cost of future generators. We distinguish these effects experimentally instead of attributing all compilation savings to adaptive physics.

The study compares sudden quenches of an eight-spin Ising ring and finite-rate ramps of an eight-spin anisotropic XY chain on a square ladder and a branching heavy-hex subgraph. These graphs separate a regime where a static embedding removes the problem from one where it cannot. A linear chain provides a further test with the same Hamiltonians. The contribution is a residual-constrained growth rule, a transparent implementation over the original AVQDS solver, and a controlled assessment of its accuracy, routing overhead, and limitations. It is not a claim of a globally optimal router or of superiority to production compilers.

1.1 Relation to variational simulation and hardware-aware compilation

Variational eigensolvers [15, 16, 17] and hardware-efficient ansatz families [18, 19, 20] provide alternatives to long coherent computations in the near-term setting [14]. Circuit structure is also a design variable in quantum optimization [21]. Chemistry-inspired constructions [22, 24] motivate adaptive selection, developed in ADAPT-VQE [23] and qubit-ADAPT-VQE [25]. These methods optimize an energy, whereas real-time variational methods follow a moving state through an evolving tangent space [26, 27]. Fast-forwarding and compressed dynamical representations address related circuit costs [28, 29, 30]; experimental variational dynamics [31] and projected variational dynamics [33] emphasize that accurate time evolution requires more than a good static ansatz. Initial-state preparation can itself use imaginary-time and adaptive approaches [41, 42, 43, 64]; that cost is distinct from the evolution circuits studied here.

Hardware awareness in adaptive selection is already established. Weaving et al. [71] bias ADAPT-VQE energy-gradient scores using an approximate subgraph-isomorphism measure. TETRIS-ADAPT-VQE [72] and AVQDS(T) [73] add disjoint supports to build denser layers. Layering and subpool exploration [74] examine commutation and screening cost, while circuit-design-informed adaptive algorithms [75] incorporate connectivity and measurement-circuit restrictions. RFG differs in using a *constraint on an actually evaluated dynamical residual improvement* and the *incremental cost of concrete persistent layouts*. It neither truncates the Hamiltonian pool nor substitutes an energy gradient for the AVQDS trial solve. It also leaves the regularized equation-of-motion solver intact, unlike the eigenvalue-truncation option studied in AVQDS(T).

Dynamic layouts and search frontiers are standard compilation ideas. SABRE [66] uses bidirectional heuristic search; architecture-independent routing and retargetable compilation are developed in tket [67, 68]; integer programming makes allocation and routing joint optimization variables [69]. Optimization-aware routing can exploit interactions between SWAPs and later gate simplification [70]. Our bounded frontier is a deliberately simple routing component, not a new generic compiler. The substantive question is whether giving its cost information to *AVQDS growth* helps beyond applying the identical routing component to an unchanged AVQDS circuit. We retain all SWAP costs and disable gate cancellation to make that comparison interpretable.

2 Methodology

2.1 Unchanged variational dynamics and candidate evaluation

Let P_j be a Pauli string in the canonical Hamiltonian pool, and let

$$|\psi(\phi)\rangle = e^{-i\phi_{p-1}P_{p-1}/2} \dots e^{-i\phi_0P_0/2} |\psi_0\rangle. \quad (1)$$

The half-angle convention is that of the supplied AVQDS implementation; it corresponds to $\theta_j = \phi_j/2$ in an $e^{-i\theta_j P_j}$ convention. Write $|d_j\rangle = \partial_j |\psi\rangle$, $q_j = \langle d_j | \psi \rangle$, and $E = \langle H \rangle$. The implementation evaluates

$$m_{jk} = \text{Re} [\langle d_j | d_k \rangle + q_j q_k], \quad v_j = \text{Im} [\langle d_j | H | \psi \rangle - q_j E], \quad (2)$$

$$u = (m + \xi I)^+ v, \quad r = \text{Var}(H) - v^T (m + \xi I)^+ v. \quad (3)$$

These are the phase-projected McLachlan equations [32, 26, 65], with the source code’s common factor of two removed. Their geometric interpretation is related to information metrics and projective-state geometry [38, 39, 40]. We use the original pseudoinverse routine with $\xi = 10^{-6}$. For the

regularized velocity u , the actual squared density-matrix residual obeys

$$L^2(u) = 2[\text{Var}(H) - 2v^T u + u^T m u] = 2r - 2\xi \|u\|_2^2 \leq 2r. \quad (4)$$

Thus the source residual is a conservative proxy once regularized. We set $r_{\text{cut}} = 5 \times 10^{-4}$, enforcing $L^2 \leq 10^{-3}$. Singular-value thresholding or a different regularization rule is not introduced.

Appending a trial rotation with $\phi_p = 0$ preserves the state but supplies the derivative $|d_p\rangle = -iP|\psi\rangle/2$. The original code augments m and v , solves Eq. (3), and computes r_P . The existing matrix blocks need no new evaluation. We reuse these trial computations and their commit bookkeeping. Growth continues until $r < r_{\text{cut}}$ and $\|u\|_\infty < 1000$; no variational rotation is removed.

To specify the underlying deterministic selector precisely, traverse the source pool in order, skipping its current last operator. Initialize $s = v^T(m + \xi I)^+ v$ and $b = \max(\|u\|_\infty, 1000)$. For a trial with projection s_P and velocity bound b_P , replace the current choice if $s_P > s + 10^{-8}$, or if $|\text{Var}(H) - s_P| < r_{\text{cut}}$ and $b_P < b$. Update s and b after each replacement. This source rule prefers better-conditioned velocities among already sufficient trials; it is not exactly a pure residual argmin. We denote its selected operator by $P_\star$ and its improvement by $\Delta_\star = r - r_{P_\star}$.

2.2 Physical layouts and an incremental routing frontier

A bijection π maps logical spin indices to physical vertices of a fixed undirected graph $G = (V, E_G)$. The Hamiltonian and observable indices always remain logical indices. Before simulation, enumerate the $8!$ maps and choose the lexicographically first maximizer of the number of Hamiltonian bonds mapped onto edges, breaking the primary tie by the smallest sum of graph distances over those bonds. The weight is one per distinct bond, so this initialization requires no future trajectory. All statically tied maps are retained for a subsequent sensitivity analysis.

Each frontier record contains its current map, cumulative SWAP count, complete gate list, and an eight-entry vector of earliest available wire depths. Initially there is one empty record. For a trial two-site operator acting at physical vertices a, b , enumerate *every shortest path* ($v_0 = a, \dots, v_d = b$). For every meeting edge $k \in \{0, \dots, d-1\}$, move the first operand along edges $0, \dots, k-1$ and the second along edges $d-1, \dots, k+1$, in that order. Exactly $d-1$ adjacent SWAPs bring the operands to edge (v_k, v_{k+1}) . Update π after each SWAP and append the physical Pauli rotation. For a single-site operator, retain the map and append its rotation. No SWAP is undone automatically.

Represent each SWAP by three CNOTs. For a two-site Pauli rotation, apply local basis changes B satisfying $B^\dagger Z B = P$, then CNOT- $R_z(\phi)$ -CNOT, then inverse basis changes. Operationally the forward basis changes are H for X and $S^\dagger$ followed by H for Y ; Z requires none. Each single-site R_x, R_y, R_z, H, S , or $S^\dagger$ costs one depth unit. A gate is placed at one plus the maximum available depth of its operands, then updates those wires. This defines total circuit depth, rather than just two-qubit depth. No cancellation or angle-dependent pruning is performed.

Deduplicate children with the same output map by lexicographic minimization of (CNOT count, maximum depth, wire-depth vector, gate list). Sort the surviving maps by (CNOT count, maximum depth, map, gate list) and keep the first $W = 16$. This beam update is applied to each admissible trial and retained only for the selected operator. The circuit used at a time point is the best current record. An expanded frontier can choose a different realization of the old logical prefix; the variational operator sequence itself remains append-only. The final map is used to interpret measurements, not restored by uncounted gates.

2.3 Residual-budgeted frontier growth

Let $S(\mathcal{F})$ be the SWAP count of the best frontier record. Let $\mathcal{F} \oplus P$ denote the candidate frontier after the extension above, and define

$$c_P = S(\mathcal{F} \oplus P) - S(\mathcal{F}),$$

$$\mathcal{A}_\rho = \left\{ P : \begin{array}{l} r - r_P \geq \rho \Delta_\star - 10^{-12}, \\ b_P < \max(1000, \|u\|_\infty) \end{array} \quad r - r_P > 10^{-8}, \right\} \cup \{P_\star\}. \quad (5)$$

We use $\rho = 0.9$. If $\Delta_\star \leq 10^{-8}$, keep $P_\star$. Otherwise find the smallest c_P in $\mathcal{A}_\rho$. Replace $P_\star$ only when this cost is strictly lower than $c_{P_\star}$; among operators at that cost, minimize $(r_P, b_P, \text{pool index})$.

Commit the selected original augmented matrices and derivative, append a zero parameter, and adopt its candidate frontier. This is the only change to AVQDS's candidate-selection decision. In particular, a route with zero SWAP overhead does not trigger a different operator merely to change the logical CNOT count.

The allowance is a constraint on measured tangent-space progress, not on exact-state fidelity. Up to the stated numerical slack, every replacement satisfies

$$r_{\text{new}} \leq (1 - \rho)r + \rho r_{P_*}. \quad (6)$$

This is a local guarantee. Neither it nor a residual threshold guarantees a particular accumulated global fidelity, nor does lower local routing cost guarantee lower final depth. Both require measurement over the whole trajectory. When every pool bond is initially native, the frontier uses no SWAPs, all routing increments are zero, and the selector recovers the original AVQDS choices exactly. This makes an embeddable graph a useful negative control.

Covariance under routing. For a logical prefix U_p and its physical realization C_p , induction over appended gates gives

$$C_p \Pi_{\pi_0} = \Pi_{\pi_p} U_p, \quad (7)$$

where Π_{π} permutes statevector tensor factors. Adjacent SWAPs change the map; conjugating a rotation by its map implements the same logical generator. The map is discrete and independent of the variational angles for each stored circuit. Consequently Eq. (7) also holds for parameter derivatives. Computing Eq. (3) in logical coordinates therefore preserves the physical variational equations exactly. Fidelity changes in RFG arise from different generator choices and finite integration error, not from noiseless SWAPs.

Integration and paired comparison. The original forward-Euler proposal is $\delta t = \min(0.05, 0.005/\|u\|_{\infty})$, clipped at the ramp boundary and final time; for zero velocity use the endpoint-clipped cap. Each method computes its own growth and proposed step. All methods in a benchmark then accept the smallest proposal and apply $\phi \leftarrow \phi + \delta t u$. This shared adaptive mesh satisfies every method's parameter-step bound without changing its EOM or reference state. The candidate decisions never access the exact trajectory.

3 Experimental setup

Physical models. We use Pauli matrices, not spin operators divided by two. The mixed-field Ising model is

$$H_I = - \sum_{i=0}^7 Z_i Z_{(i+1) \bmod 8} - 2 \sum_{i=0}^7 X_i + h_z \sum_{i=0}^7 Z_i. \quad (8)$$

The initial state is $|0\rangle^{\otimes 8}$, the full Hamiltonian is switched on at $t = 0$, and evolution ends at $t = 3$. We study $h_z = 0$ and 0.5 using respectively the 16- and 24-element Hamiltonian pools. The return probability $\mathcal{L}(t) = |\langle \psi_0 | \psi(t) \rangle|^2$ probes the dynamics; its relevance to quenches and dynamical transitions is well established [58, 59].

For the open Lieb–Schultz–Mattis chain [34],

$$H_{XY}(t) = - \sum_{i=0}^6 [(1 + \gamma(t))X_i X_{i+1} + (1 - \gamma(t))Y_i Y_{i+1}] + h_z \sum_{i=0}^7 Z_i, \quad (9)$$

with $\gamma(t) = 1 - 2t/3$ for $0 \leq t \leq 3$ and $\gamma(t) = -1$ thereafter. Fields are $h_z = -0.7$ and 1.6 . Exact diagonalization supplies the initial ground state; dynamics continues to $t = 6$. The pool contains eight Z_i and fourteen nearest-neighbor XX/YY terms, including those whose coefficient vanishes at a ramp endpoint. This finite-rate passage between anisotropic ferromagnetic regimes is distinct from an adiabatic ground-state calculation; quantum critical dynamics supplies useful physical context [35]. We measure energy and XX, YY correlations on logical pairs $(0, 1)$ and $(0, 7)$. No initial-state preparation circuit is charged to any method: costs refer to evolution conditional on the common prescribed state, not to an end-to-end ground-state preparation experiment.

Device graphs. The square ladder has edges

$$E_{\square} = \{01, 12, 23, 45, 56, 67, 04, 15, 26, 37\}, \quad E_{\text{hex}} = \{01, 12, 13, 35, 45, 56, 67\}.$$

Here ab denotes an undirected pair. We also test $E_{\text{line}} = \{01, 12, 23, 34, 45, 56, 67\}$. These are hardware graphs; neither changes the physical Hamiltonian. The square initial map is $(0, 1, 2, 3, 7, 6, 5, 4)$, read as physical positions of logical sites $0, \dots, 7$. The prescribed deterministic rule chooses the identity map on the other graphs. The square maps all bonds natively. On the heavy-hex graph, only five distinct Hamiltonian bonds can be native for either model, as proved below.

Baselines and ablations. Unconstrained AVQDS uses the directly imported source growth routine. The smart-embedding naive baseline compiles that *same recorded circuit*: for every nonlocal two-site rotation take the lexicographically first shortest path, move its first listed operand toward the second until adjacent, append the rotation, and retain the resulting map. This permits layout reuse. We additionally evaluate shortest-path routing with restoration after each rotation, since neither deterministic rule is guaranteed to dominate the other. The routing-only controls apply our width-one and width-sixteen frontiers to the unchanged unconstrained circuit. The growth sensitivity changes only ρ from 0.9 to 0.5 on heavy-hex. All these settings were specified before inspecting complete trajectories. All four physical configurations are reported; none is selected on performance.

Numerics, validation, and cost accounting. The original source’s `one_step`, phase projection, Tikhonov pseudoinverse, augmented trial calculations, and Euler update are retained. Exact signed-permutation Pauli actions and batched tangent propagation replace dense repeated exponentiation, with numerical regression against the original implementation. Compatibility fixes restore removed NumPy type aliases and correct a Z-operator’s metadata label; pool matrices are checked unchanged. A new driver applies the full Ising quench at the correct initial time, timestamps the updated state correctly, and lands exactly on each endpoint. These are shared baseline adaptations, not topology-specific changes.

For constant Hamiltonians, the reference uses a spectral exponential and is independently checked against `scipy.linalg.expm`. For the XY ramp, two independently refined DOP853 meshes use maximum steps 0.02 and 0.01, relative tolerances 2×10^{-11} and 2×10^{-13} , and absolute tolerances one twentieth as large. Independent midpoint propagators $\exp[-iH(t + \delta t/2)\delta t]$, evaluated by dense `scipy.linalg.expm` with step caps 0.002 and 0.001 and clipped at all output times, check this reference by a different method. After $t = 3$, a spectral exponential is cross-checked by `expm`. Numerical integration and conditioning are treated as separate issues [49, 50]; no exact state is passed to `ansatz` growth.

Every time point records fidelity, infidelity, parameters, residual, time-step proposal, energy, return probability, and correlations. For an accepted step, the residual is its post-growth, pre-Euler value; the associated state and parameters are its updated endpoint values. The initial empty-ansatz row records the initial energy variance before growth. Primary physical circuits are replayed from the mapped reference state at *every* time point. Every CNOT pair is separately checked against its graph; SWAPs are already decomposed. Saved circuit definitions place the source logical sequence beside its routed gate list. CNOT counts are $2N_{\text{two-site}} + 3N_{\text{SWAP}}$, consistently with the standard rotation decomposition [37]. Depth includes all specified single-qubit gates. Final output permutations are interpreted classically for both routed methods. We do not include shot noise, calibration-dependent gate durations, or noise channels.

There is one deterministic trajectory per stated configuration, with all methods paired by exact initial-state and time-array hashes. We report all trajectories and all tied-embedding controls, without treating time points as independent statistical replicates. Four benchmark groups run as separate processes with one BLAS thread each. Software versions and measured costs are retained in run manifests. Timings cover the evolution kernels, recorded embedding enumeration, and any external compilation; reference evolution, audit replay, and file serialization are reported separately from algorithmic work. Pool construction and common reference-state preparation are excluded. This is a timing observation on the present CPU environment, not a hardware-independent speed claim.

4 Experimental results

4.1 Accuracy and executable circuit cost

Tables 1 and 2 give every required final-time comparison. The unconstrained Ising runs reach fidelities above 99.96%, comfortably reproducing the high-accuracy regime of AVQDS. Their CNOT counts differ from the published examples, as expected from the explicitly stated residual convention, half-angle step restriction, shared mesh, and numerical ordering. The XY trajectories remain accurate throughout the ramp and subsequent evolution, with small circuit growth after the ramp. These are newly executed baseline trajectories, not the original article’s reference data.

Across both device graphs, the lowest final RFG fidelity is 99.96035%. The largest loss relative to its paired unconstrained trajectory is 0.00638 percentage points, well below a one-percentage-point accuracy allowance. Routing the unchanged baseline preserves its state to floating-point accuracy. On the square ladder, RFG also preserves all growth decisions, parameters, and circuit costs. On heavy-hex, fidelity need not be identical because the generator sequence changes. The $h_z = 1.6$ XY trajectory slightly improves, while the other three final fidelities slightly decrease; there is no uniform accuracy improvement claim.

Table 1: Ising final-time results at $t = 3$. Naive routing uses the unchanged AVQDS circuit and an optimal static bond-coverage map. All single-qubit gates and all three CNOTs per SWAP contribute to depth D .

Method	Graph	$h_z = 0$			$h_z = 0.5$		
		f (%)	N_{cx}	D	f (%)	N_{cx}	D
AVQDS	All-to-all	99.98241	140	106	99.96399	232	171
Naive	Square	99.98241	140	106	99.96399	232	171
Naive	Heavy-hex	99.98241	398	322	99.96399	670	510
RFG	Square	99.98241	140	106	99.96399	232	171
RFG	Heavy-hex	99.98146	205	146	99.96035	418	296

Table 2: XY final-time results at $t = 6$. Naive routing uses the unchanged AVQDS circuit and an optimal static bond-coverage map. All single-qubit gates and all three CNOTs per SWAP contribute to depth D .

Method	Graph	$h_z = -0.7$			$h_z = 1.6$		
		f (%)	N_{cx}	D	f (%)	N_{cx}	D
AVQDS	All-to-all	99.98249	118	175	99.96141	118	200
Naive	Square	99.98249	118	175	99.96141	118	200
Naive	Heavy-hex	99.98249	229	316	99.96141	244	330
RFG	Square	99.98249	118	175	99.96141	118	200
RFG	Heavy-hex	99.97611	166	263	99.97371	164	251

Table 3: Overhead relative to the same unconstrained run. Savings are total CNOT reductions relative to naive routing; they include changes in logical circuit size.

Case	Graph	Naive / AVQDS		RFG / AVQDS		Saving (%)
		CNOT	Depth	CNOT	Depth	
Ising 0	Square	1.000	1.000	1.000	1.000	0.0
Ising 0	Heavy-hex	2.843	3.038	1.464	1.377	48.5
Ising 0.5	Square	1.000	1.000	1.000	1.000	0.0
Ising 0.5	Heavy-hex	2.888	2.982	1.802	1.731	37.6
XY -0.7	Square	1.000	1.000	1.000	1.000	0.0
XY -0.7	Heavy-hex	1.941	1.806	1.407	1.503	27.5
XY 1.6	Square	1.000	1.000	1.000	1.000	0.0
XY 1.6	Heavy-hex	2.068	1.650	1.390	1.255	32.8

The heavy-hex cost reduction is substantial even against persistent shortest-path routing. RFG reduces final CNOT counts by 27.5–48.5%, with an unweighted mean reduction of 36.6% across the four

Table 4: Parameter counts and heavy-hex routing. Naive routing preserves the AVQDS parameter count. The final column counts growth decisions changed by RFG.

Case	N_ϕ			SWAPs, heavy-hex		Choices changed
	AVQDS	Square	Hex	Naive	RFG	
Ising 0	143	143	127	86	31	44
Ising 0.5	301	301	310	146	68	63
XY -0.7	72	72	81	37	16	13
XY 1.6	73	73	76	42	14	14

problems. The mean per-problem CNOT multiplier falls from 2.435 to 1.516, and the mean depth multiplier from 2.369 to 1.467 (Table 3). These are arithmetic means of paired ratios, rather than ratios of pooled gate totals. The square multipliers are exactly one for both methods. Parameter counts and explicit SWAP counts in Table 4 show that reducing routing does not require uniformly fewer parameters.

4.2 Dynamical observables and independent validation

Figures 1 and 2 resolve accuracy and cost over the entire trajectories. The return probabilities in Fig. 3 and the XY energy and correlations in Figs. 4–5 follow the exact dynamics closely. The ramp transfers the dominant nearest-neighbor correlation between spin components; the long-range correlations retain the effects of the finite ramp speed. Post-ramp energy is approximately conserved, with a small variational drift. These checks constrain the interpretation of high fidelity: the agreement is also visible in physical observables, while the remaining observable errors are nonzero.

The maximum difference between the two refined XY reference meshes is 1.24×10^{-10} in statevector norm over all recorded times. The independent midpoint matrix-exponential calculations improve with mesh refinement in both cases; the finer cap agrees with DOP853 within 3.95×10^{-7} . Physical replay of primary circuits at every step, and final replay of all compilation controls, gives a maximum logical/physical vector difference of 2.23×10^{-14} . The audits check 49, 453, 552 CNOT instances across time points and methods, with zero edge-set violations. Repeated occurrences of a prefix at different times count separately in that audit total; it is a verification workload, not the cost of one experimental circuit. Complete per-step pair lists and flags are stored separately for each topology.

5 Analysis

5.1 Mechanism analysis: routing reuse and tangent-space choices

The graph obstruction is sharper than edge counting. The heavy-hex subgraph is a tree with two nonadjacent degree-three vertices. The native bonds of an embedded path or cycle form a subgraph of maximum degree two. At least one incident edge must be omitted at each branching vertex; because those vertices are nonadjacent, the omissions are distinct. At most five of the seven tree edges can therefore host distinct native interaction bonds. The identity placement attains five for both physical problems. Consequently the ring has at least three nonnative bonds, and the open chain at least two. In contrast, the square ladder has the Hamiltonian cycle used by our embedding, so its entire Hamiltonian pool is native.

There is also a distance obstruction. A tour visiting all vertices of an eight-vertex tree and returning to its start must traverse every edge at least twice, giving total distance at least 14. An open tour can save at most the tree diameter, which is 5 here, giving a minimum distance sum of 9. Our coverage-optimal embeddings attain both lower bounds. For a complete sweep over ring ZZ bonds, independent shortest-path routing with restoration would therefore require at least $16 + 6(14 - 8) = 52$ CNOTs; for one sweep over both open-chain XX and YY bonds it requires $28 + 12(9 - 7) = 52$. These are structural benchmarks for *restoring full sweeps*, not lower bounds on a learned variational circuit. They explain why optimizing only the initial placement cannot remove the heavy-hex penalty.

Which component produces the reduction? Table 5 and Fig. 6 separate compilation from changed growth. On the integrable Ising trajectory, naive routing uses 398 CNOTs. Applying the width-sixteen frontier to its unchanged operator sequence reduces this to 278; permitting residual-budgeted growth

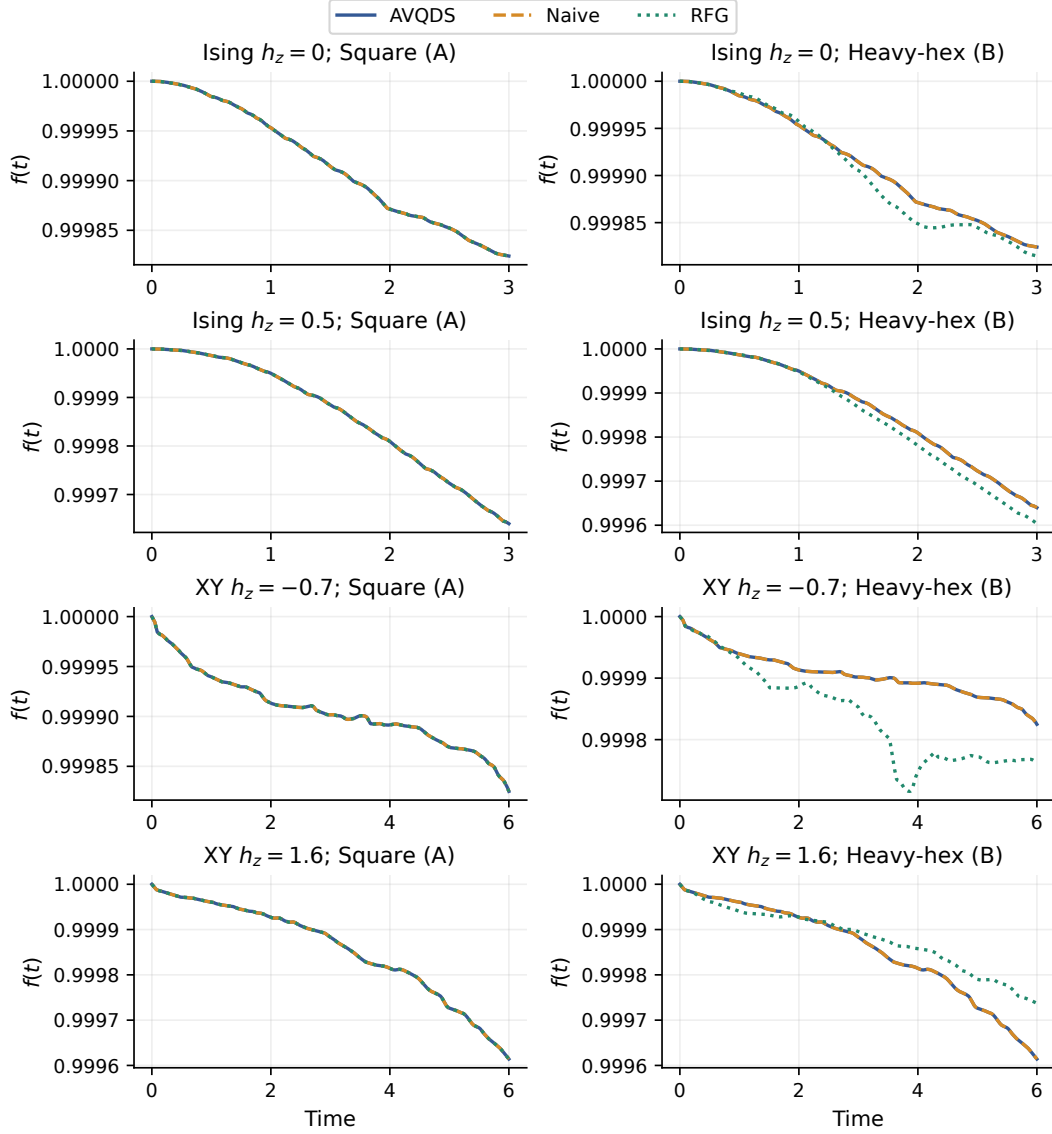


Figure 1: Fidelity at every accepted time point for all four problems and both device graphs. All three methods use the same reference state and adaptive time mesh. Square-lattice curves coincide; naive routed curves coincide with unconstrained AVQDS on either graph. Heavy-hex RFG changes the growth sequence, producing the small resolved differences.

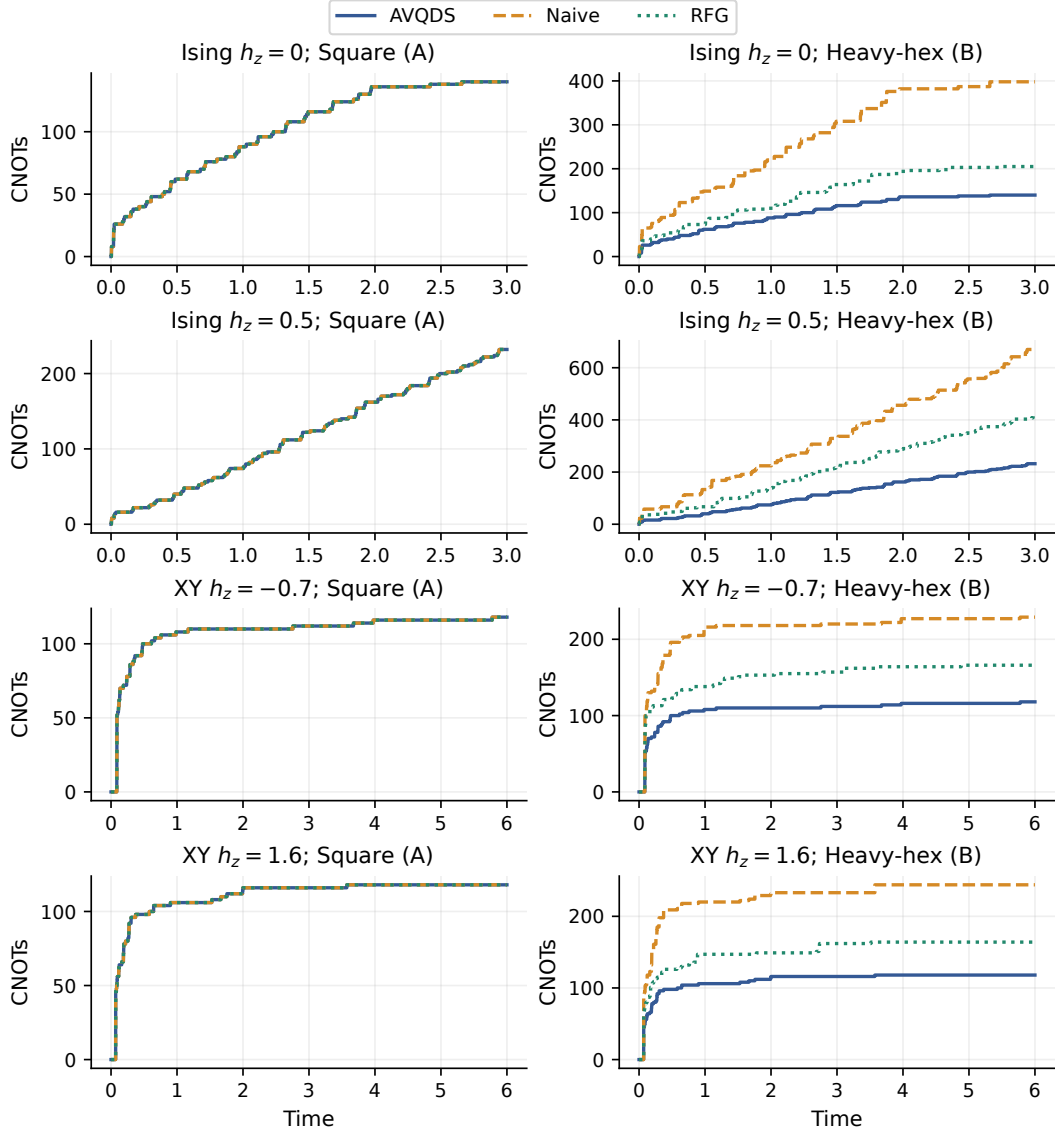


Figure 2: Executable CNOT count versus time, including three CNOTs for every SWAP. The Ising nonintegrable trajectory continues to grow over the simulated interval; XY counts nearly saturate. The square-lattice overlap is the expected zero-overhead control, not an omitted routing cost.

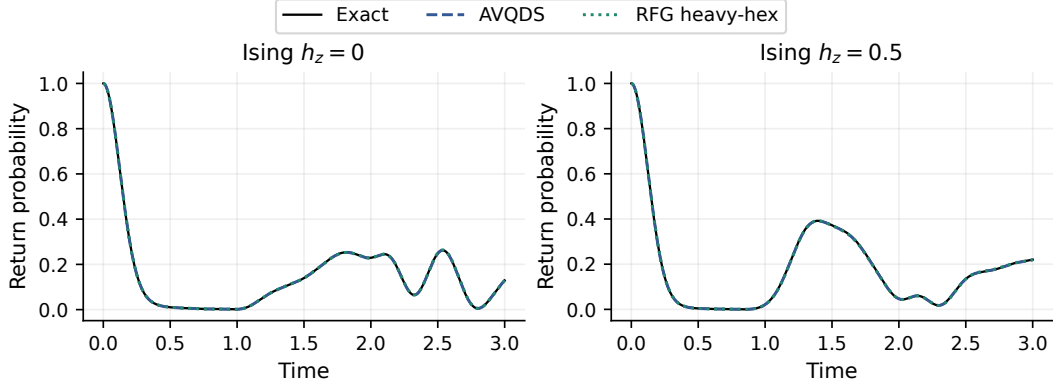


Figure 3: Ising Loschmidt echoes. Exact, unconstrained, and heavy-hex RFG trajectories are overlaid. Square RFG and naive routing reproduce the unconstrained state and are omitted here to avoid identical additional curves.

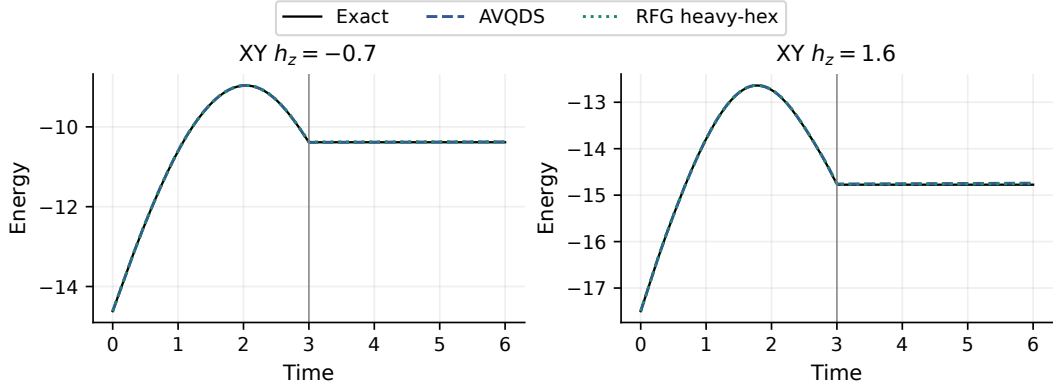


Figure 4: XY-chain energy through the linear ramp and post-ramp evolution. The vertical line marks the end of the ramp. The plotted energies use the original logical Hamiltonian, with final layout permutations accounted for.

further reduces it to 205. The latter change combines fewer logical rotations (140 to 112 logical CNOTs) with fewer SWAPs (46 to 31 relative to routing alone). In the nonintegrable Ising case the corresponding sequence is $670 \rightarrow 517 \rightarrow 418$ CNOTs. Thus the growth rule contributes beyond ordinary routing, rather than merely renaming a compiler improvement.

The XY cases give a more discriminating result. For $h_z = -0.7$, RFG has the same 118 logical CNOTs as the baseline but changes how they fit onto the device: the width-sixteen routing-only control uses 20 SWAPs, versus 16 for RFG. It also introduces additional single-site rotations, raising the parameter count from 72 to 81. For $h_z = 1.6$, logical CNOTs change from 118 to 122, while the routing-only SWAP count drops from 22 to 14. Hardware savings can therefore compensate for a slightly larger logical ansatz. Figure 7 shows the actual physical edges used, including the routing traffic through the branching region.

Dependence on the smart embedding. An arbitrary poor initial placement could exaggerate a routing improvement. We instead enumerate all static maps and use the same predeclared coverage/distance rule for every method. Table 6 further compiles every tied heavy-hex embedding of the baseline, with all individual outcomes retained. RFG’s CNOT count is lower even than the best naive count among those tied maps in each problem. Table 7 completes the restoring and line-graph controls. On the square graph all tied native embeddings have the same costs. This checks the sensitivity of the *naive comparator*; it does not establish that RFG itself is insensitive to changing its initial map, since full RFG trajectories were run only from the prescribed map.

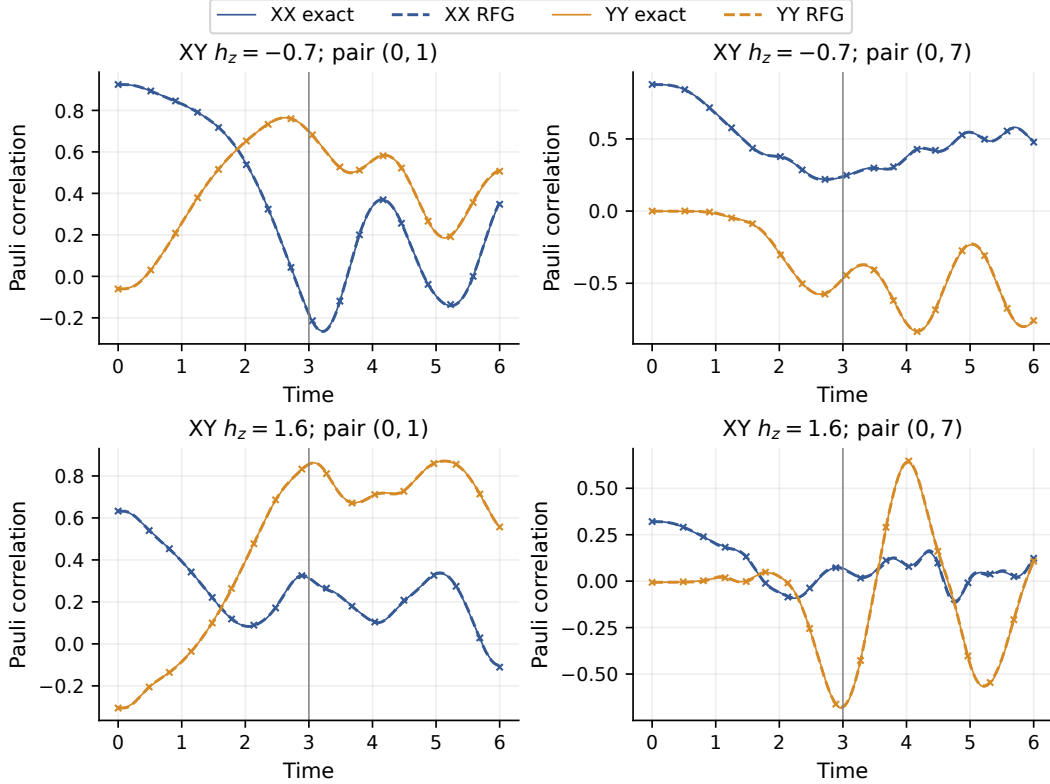


Figure 5: XY-chain Pauli correlations for nearest and end-to-end logical pairs. Solid lines are exact; dashed lines are heavy-hex RFG; crosses mark unconstrained AVQDS at selected times. Colors distinguish XX and YY . No factor of four for spin-1/2 operators has been introduced.

Table 5: Complete heavy-hex ablation: each entry is CNOT count / total depth. Routing-only controls preserve the source AVQDS state exactly.

Construction	Ising 0	Ising 0.5	XY -0.7	XY 1.6
Naive, persistent	398 / 322	670 / 510	229 / 316	244 / 330
Naive, restoring	428 / 368	718 / 591	226 / 299	214 / 285
Routing only, $W = 1$	347 / 241	784 / 488	178 / 250	184 / 262
Routing only, $W = 16$	278 / 203	517 / 365	178 / 250	184 / 261
RFG, $\rho = 0.9$	205 / 146	418 / 296	166 / 263	164 / 251
RFG, $\rho = 0.5$	213 / 148	383 / 277	139 / 228	130 / 164

5.2 Failure analysis: local savings do not define a global optimum

RFG is not a uniformly better circuit constructor along every axis. Its square-lattice result is a tie with smart naive routing; once every Hamiltonian bond is native, there is no SWAP overhead for this method to remove. This is an intentional boundary of the contribution. On heavy-hex it still incurs nontrivial overhead over the unconstrained circuit in every primary case. The nonintegrable Ising trajectory has the largest primary CNOT multiplier, while the longer XY evolution does not imply the largest cost: expressibility saturation and routing structure matter more than elapsed physical time alone.

The clearest objective mismatch is the $h_z = -0.7$ XY case. RFG reduces CNOTs from 178 to 166 relative to width-sixteen routing alone, but increases depth from 250 to 263. Its extra single-site rotations and changed dependency pattern offset the saved SWAPs in the critical path. Equation (5) prioritizes SWAP count; depth is only a secondary frontier criterion. It cannot guarantee a depth improvement over an alternative generator sequence. Similarly, the width-one routing control

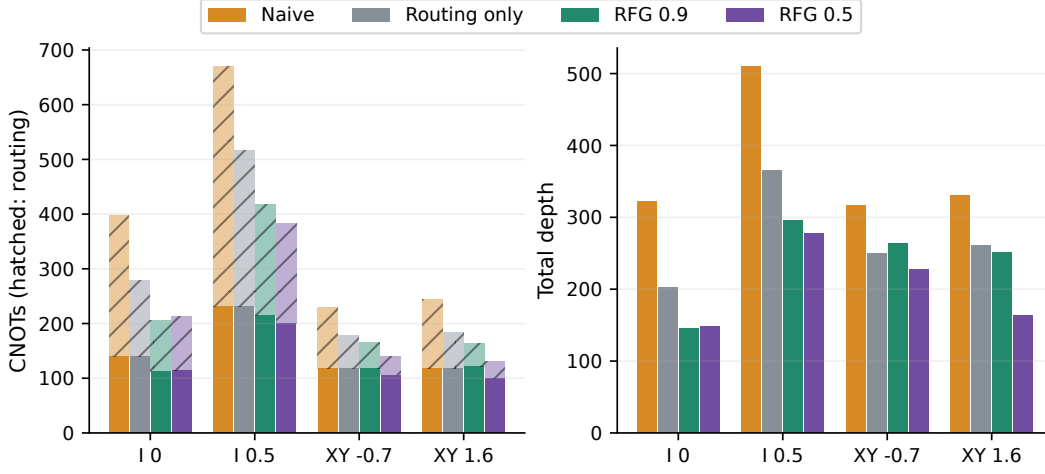


Figure 6: Heavy-hex mechanism and sensitivity. Hatched CNOT portions are routing costs; solid solid portions are variational two-site rotations. The depth panel includes basis changes and single-site rotations, so it need not track the CNOT ordering. The stronger residual allowance helps several cases but does not dominate the primary setting.

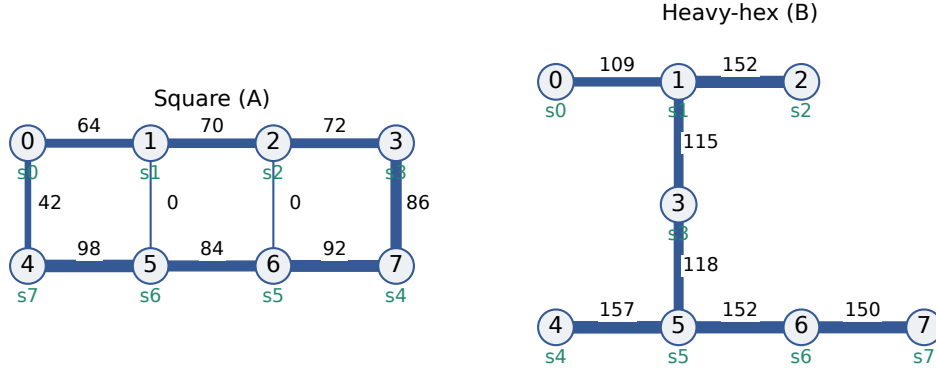


Figure 7: Topology compliance and initial embedding. Node numbers are physical vertices; green labels are their initial logical sites. Edge labels and widths give the sum of final RFG CNOT counts over the four problems. Only allowed edges appear. These aggregate edge counts visualize traffic, not a simultaneous execution schedule.

performs poorly on nonintegrable Ising: 784 CNOTs exceed the naive count of 670. Greedily selecting a meeting edge can sacrifice a layout that would serve later rotations.

The predeclared $\rho = 0.5$ sensitivity exposes a separate trade-off (Tables 5 and 8). It improves heavy-hex CNOTs for nonintegrable Ising and both XY cases, but worsens integrable Ising from 205 to 213, also lowering its fidelity. The primary $\rho = 0.9$ choice is therefore not a retrospectively optimized hyperparameter. A larger residual allowance can unlock cheap useful directions, but can also force extra work later. In the $h_z = 1.6$ XY case the stronger allowance even lowers total depth below the unconstrained baseline; that is a change in the learned ansatz and does not mean a fixed circuit can be routed at negative cost.

Persistent naive routing also need not beat restoration: both XY cases are cheaper with restoration, whereas the Ising cases favor persistence. Both strategies are shown in Table 5; primary RFG improves on both in all four cases. Omitting the restoring control would overstate how universal the

Table 6: Exhaustive static-embedding sensitivity on heavy-hex. All maps tied under the stated coverage/distance criterion are compiled by naive persistent routing. Summaries show minimum / median / maximum; RFG uses the preselected lexicographic map.

Case	Maps	Naive CNOT range	Naive depth range	RFG CNOT
Ising 0	320	344 / 398 / 440	247 / 305 / 344	205
Ising 0.5	320	616 / 715 / 799	462 / 537 / 619	418
XY -0.7	16	226 / 241 / 262	313 / 336.5 / 359	166
XY 1.6	16	238 / 281.5 / 322	315 / 367.5 / 401	164

Table 7: Remaining exhaustive embedding controls: minimum / median / maximum over every tied map, completing Table 6. On the square graph all 16 ring maps and all 28 open-chain maps have zero routing overhead for both controls.

Case	Graph	Routing	Maps	CNOT range	Depth range
Ising 0	Hex	Restore	320	374 / 458 / 524	301 / 388.5 / 476
Ising 0	Line	Persist	16	332 / 384.5 / 437	204 / 229.5 / 248
Ising 0	Line	Restore	16	356 / 446 / 536	324 / 406 / 506
Ising 0.5	Hex	Restore	320	688 / 760 / 802	531 / 625 / 722
Ising 0.5	Line	Persist	16	727 / 779.5 / 832	402 / 436.5 / 473
Ising 0.5	Line	Restore	16	664 / 772 / 808	603 / 706.5 / 754
XY -0.7	Hex	Restore	16	208 / 217 / 244	277 / 288.5 / 299
XY -0.7	Line	Persist	2	118 / 118 / 118	175 / 175 / 175
XY -0.7	Line	Restore	2	118 / 118 / 118	175 / 175 / 175
XY 1.6	Hex	Restore	16	208 / 217 / 244	281 / 297 / 307
XY 1.6	Line	Persist	2	118 / 118 / 118	200 / 200 / 200
XY 1.6	Line	Restore	2	118 / 118 / 118	200 / 200 / 200

persistent-layout heuristic is. All non-wins and the backend regression-fixture correction are recorded in the accompanying attempts log.

Table 8: All stronger-bias ($\rho = 0.5$) trajectories on heavy-hex. CNOTs and depth are in Table 5; these rows complete their accuracy, parameter, and timing account.

Case	f (%)	N_ϕ	SWAPs	Time (s)
Ising 0	99.97664	126	33	26.26
Ising 0.5	99.96160	305	61	81.66
XY -0.7	99.98735	74	11	31.47
XY 1.6	99.96572	73	10	31.55

5.3 Computational cost trade-off

The source solver dominates classical work (Table 9). Relative to unconstrained AVQDS, heavy-hex RFG’s measured algorithmic time ranges from $0.787\times$ to $1.163\times$. The routing-frontier work itself occupies only 48.6–336.3 ms per trajectory. The longer runtime in the negative-field XY case follows its larger parameter count; the integrable Ising run becomes faster because its ansatz is smaller. Small timing differences between mathematically identical square and unconstrained trajectories should be read as runtime variability, not as a solver speedup.

For Hilbert-space dimension $d = 2^n$, p parameters, and pool size K , our exact Pauli/tangent representation requires $O(dp^2)$ work for tangent propagation and metric formation. The unchanged inverse costs $O(p^3)$, and a growth event can call it for each trial, costing $O(Kp^3)$ in the direct implementation. The frontier adds a bounded classical search over W maps, shortest paths, and meeting edges. Its cost is small at $n = 8$, but enumeration of all initial maps has factorial scaling. A large-system implementation needs a heuristic placement or a separate allocation optimizer; the present exhaustive placement is an auditable small-system control, not a scalable algorithmic claim. In addition, storing full gate lists rather than backpointers is convenient here but not an asymptotically optimized design.

The trial quantities are the same expectation values used by ordinary variational dynamics and can be obtained by the standard measurement constructions [44, 45, 46]. RFG evaluates the same complete pool at an expansion, then performs classical comparisons on those scores. It does not introduce additional observables per trial, although different parameter counts and numbers of expansions can change total measurement work. Finite-shot allocation and metric-estimation costs remain separate concerns [51, 54]; optimization-based alternatives to the inverse [52] were not substituted. The simulation uses the supplied QuTiP-based model construction [53], NumPy, and SciPy, with exact statevectors throughout.

Table 9: Measured classical cost on paired meshes. Evolution plus mapping/compilation is in seconds; the last column is the routing-frontier portion of RFG evolution in milliseconds. Common state preparation, pool construction, exact evolution, and verification are excluded.

Case	AVQDS	Naive A	Naive B	RFG A	RFG B	B/base	Route (ms)
Ising 0	33.17	33.21	33.22	33.05	26.09	0.787	84.2
Ising 0.5	78.93	78.99	79.03	79.15	84.13	1.066	336.3
XY -0.7	31.59	31.61	31.62	31.22	36.74	1.163	48.6
XY 1.6	32.62	32.64	32.65	32.24	32.97	1.011	81.3

5.4 Generalization and scope

The linear-chain results in Table 10 preserve the same pattern without changing any physical benchmark or hyperparameter. Both XY chains embed natively and recover the original ansatz. The periodic Ising ring does not; RFG reduces routing overhead there while preserving high fidelity. All line ablations are in Table 11. Integrable Ising again illustrates the depth trade-off: RFG uses fewer CNOTs than width-sixteen routing alone but increases depth from 177 to 185. This shows that the rule is portable across graphs, but does not establish scaling with system size.

Entanglement growth and the distinction between compact quantum circuits and classical state representations [60, 61, 62, 63] motivate adaptive dynamics, but our exact eight-qubit simulations do not demonstrate a quantum computational advantage. Nor do they test barren-plateau mitigation [55, 56, 57]: parameters are advanced by a variational EOM, not by a generic energy-training optimizer. Symmetry-based pool reductions [47, 48] could be combined with connectivity-aware growth, provided the original physical dynamics remains representable. That combination is outside these experiments.

The fixed Hamiltonian pool, noiseless gates, and conditional LSM initialization are material limits. Routing gains alone do not establish improved fidelity on a noisy processor, especially when single-qubit gates or idle durations affect error. No Trotter advantage is measured here; product-formula simulation [36] is background rather than a silently assumed comparator. Comparisons against SABRE, tket, more expressive pools, additional system sizes, and noisy or finite-shot execution would answer different questions. The present evidence isolates a narrower statement: residual-constrained generator choices improve the cost of sparse-graph AVQDS beyond the same small routing search applied after growth.

Table 10: Generalization to an eight-qubit line with the same four physical problems. Paired entries are CNOT count / depth; final columns are RFG multipliers over unconstrained AVQDS.

Case	f (%)	Naive	RFG	N_ϕ	CNOT $\times$	Depth $\times$
Ising 0	99.97702	437 / 248	256 / 185	139	1.829	1.745
Ising 0.5	99.96047	790 / 443	417 / 271	303	1.797	1.585
XY -0.7	99.98249	118 / 175	118 / 175	72	1.000	1.000
XY 1.6	99.96141	118 / 200	118 / 200	73	1.000	1.000

Table 11: All linear-chain routing ablations, with CNOT count / depth in each cell. On the square ladder every corresponding control exactly equals the unconstrained circuit cost, so there are no additional distinct square outcomes.

Construction	Ising 0	Ising 0.5	XY -0.7	XY 1.6
Naive, persistent	437 / 248	790 / 443	118 / 175	118 / 200
Naive, restoring	428 / 398	700 / 623	118 / 175	118 / 200
Routing only, $W = 1$	359 / 195	580 / 291	118 / 175	118 / 200
Routing only, $W = 16$	272 / 177	487 / 305	118 / 175	118 / 200
RFG	256 / 185	417 / 271	118 / 175	118 / 200

6 Conclusion

Residual-budgeted frontier growth couples topology cost to the construction of an adaptive dynamical ansatz while retaining the original McLachlan solver. It uses a local residual-improvement allowance to choose between useful generators and carries explicit physical layouts across expansions. The resulting circuits obey the fixed connectivity exactly, with every SWAP charged to their cost.

The experiments distinguish two structural regimes. A square-lattice cycle makes both Hamiltonian pools native and yields unit overhead. On the branching heavy-hex graph, where no static embedding can do so, RFG lowers the mean CNOT multiplier from 2.435 to 1.516 and the mean depth multiplier from 2.369 to 1.467, with final fidelity at least 99.96035%. Routing-only controls show that changed ansatz growth contributes to these savings. The depth regression in one XY case and the mixed response to a stronger cost bias show why local SWAP savings are not a global optimality principle. Hardware-aware adaptive dynamics benefits from treating expressibility and placement together, while measuring their trade-offs separately.

Reproducibility and data availability. The accompanying artifact contains executable drivers in `code/`, full state and parameter trajectories, per-step physical gate audits, all embedding controls, and machine-readable summaries in `logs/`. Tables, figures, and numerical macros are generated from those records. `codebase_notes.md` identifies reused source routines and compatibility adaptations; `attempts_log.md` records every attempted configuration and non-win. The reference implementation remains an explicit local dependency rather than being replaced by an undisclosed implementation. No incomplete scientific experiment is relegated to an appendix.

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